

Airbnb: Delivery 1

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1 Introduction

In the present document, it will be boarded the system description of Airbnb, which will the appropriate analysis of its business problem, which actors take part of it, describe its main Use Cases, and define the boundaries within the system, i.e., what are the limits of the designed and development of the system.

Then, it will be redacted and described the Functional and Non-Functional Requirements that the system accomplishes in a real-life situation. The values of the Non-Functional Requirements, will be defined using Airbnb Workload Model, which will appear in its own section in the document.

The boundaries and limits, will also specify how will the real Non-Functional Requirements be taken into account for its simulation.

2 System Description

In the present section, there will be a description of Airbnb system, which will take further the business problem it was designed to solve, as well as its main actors, and how they interact within the system described in the corresponding Use Cases.

Finally, there will be defined the System Boundaries, i.e, what is inside of the scope of our system and what is explicitly outside of it.

2.1 Business Problem

The problem that Airbnb solves is referred to the fragmented, inefficient and uncertain access to short-term accommodations. It gives people who are in need of a place to stay, and people with those accommodations or properties available. The hosts might find problems finding or reaching potential guests, managing bookings and monetizing after all the stay, and guests might struggle finding said place.

Airbnb provides a platform in which guests are benefited with a large range and access of accommodation options, and hosts can get additional income and a wider spectre of possible guests around the world.

Its operational challenge at scale, comes when one must maintain availability accuracy (concurrency), transaction integrity, trust, and service reliability across a global two-sided marketplace. Also, there might be some problems taking compliance into account.

2.2 Actors

Taking actors, as the human users and the external services that are interacting with our application, the following schemas describe their role on it, how much they interact with the application, and what could they expect.

Human users

- **Visitors:** They browse listing and explore accommodations without having an account. Its interaction frequency is high, as they would use the main requirements of the application without counting logged in users. They would expect fast searches, useful filters, accurate prices, clear property descriptions, the map-based discovery and a simple registration process.
- **Guests:**
- **Hosts:**
- **Co-hosts and property managers:**

- Auditors:
- Trust and safety specialists:
- Administrators:
- Customer Support:

External Services

- Geolocation / Maps API:
- Payment Gateway:
- Authentication Service:
- Identity Verification Service:
- Notification Service:
- Fraud Detection Service and Risk Management: